



Transcriptomic and Proteomic of Gastrocnemius Muscle in Peripheral Artery Disease

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BACKGROUND: Few effective therapies exist to improve lower extremity muscle pathology and mobility loss due to peripheral artery disease (PAD), in part because mechanisms associated with functional impairment remain unclear.

METHODS: To better understand mechanisms of muscle impairment in PAD, we performed in-depth transcriptomic and proteomic analyses on gastrocnemius muscle biopsies from 31 PAD participants (mean age, 69.9 years) and 29 age- and sex-matched non-PAD controls (mean age, 70.0 years) free of diabetes or limb-threatening ischemia.

RESULTS: Transcriptomic and proteomic analyses suggested activation of hypoxia-compensatory mechanisms in PAD muscle, including inflammation, fibrosis, apoptosis, angiogenesis, unfolded protein response, and nerve and muscle repair. Stoichiometric proportions of mitochondrial respiratory proteins were aberrant in PAD compared to non-PAD, suggesting that respiratory proteins not in complete functional units are not removed by mitophagy, likely contributing to abnormal mitochondrial activity. Supporting this hypothesis, greater mitochondrial respiratory protein abundance was significantly associated with greater complex II and complex IV respiratory activity in non-PAD but not in PAD. Rate-limiting glycolytic enzymes, such as hexokinase and pyruvate kinase, were less abundant in muscle of people with PAD compared with non-PAD participants, suggesting diminished glucose metabolism.

CONCLUSIONS: In PAD muscle, hypoxia induces accumulation of mitochondria respiratory proteins, reduced activity of rate-limiting glycolytic enzymes, and an enhanced integrated stress response that modulates protein translation. These mechanisms may serve as targets for disease modification.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: biomarkers ■ mitochondria ■ mitophagy ■ proteomics ■ unfolded protein response

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Peripheral artery disease (PAD) is caused by stenotic atherosclerotic plaques in lower extremity arteries that reduce lower extremity tissue perfusion. During walking, increased oxygen demand in lower extremity causes an imbalance between oxygen requirement and availability that results in ischemia.¹ Chronic hypoxia and ischemia-reperfusion episodes damage muscle and may induce compensatory responses, such as upregulation of angiogenesis and mitochondrial biogenesis.^{2–4} Cilostazol

is the only drug recommended by clinical guidelines for PAD but benefits are modest.⁵ A better understanding of mechanisms associated with muscle lower extremity ischemia may reveal new candidate intervention target to prevent or ameliorate PAD-induced muscle pathology and mobility loss. This cross-sectional study compared the transcriptome and proteome of gastrocnemius muscle between patients with PAD and non-PAD controls to identify novel mechanisms that, if confirmed by

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Novelty and Significance

What Is Known?

- Peripheral artery disease (PAD) reduces blood flow to the legs, causing muscle damage and walking impairment.
- Few therapies to improve functional performance are available for PAD, and identifying biological pathways of skeletal muscle damage in PAD may provide targets for novel therapies.

What New Information Does This Article Provide?

- Both mRNA and proteins were characterized in lower extremity muscle biopsies from 31 people with PAD and compared to mRNA and proteins in lower extremity muscle biopsies from 29 people without PAD.
- Compared with the muscle from people without PAD, muscle from the people with PAD showed evidence of inflammation, mitochondrial dysfunction, and impaired glycolysis along with compensatory responses and evidence of proteins that repair damaged DNA, proteins, and muscle fibers.
- In people with PAD, mitochondrial proteins were present in disproportionate quantities, indicating over abundance of some mitochondrial proteins relative to others, consistent with the presence of mitochondrial fragments not removed by mitophagy.

PAD is caused by atherosclerotic plaques that reduce lower extremity blood flow and oxygen delivery to the lower extremities, resulting in lower extremity skeletal muscle damage and impaired walking performance. No highly effective drug treatment exists for PAD. To identify biologic pathways of skeletal muscle damage in people with PAD that may serve as therapeutic targets for novel interventions, we performed extensive analysis of mRNA and proteins in lower extremity muscle biopsies from 31 people with PAD and 29 controls without PAD. Compared to muscle from people without PAD, PAD muscle showed evidence of inflammation, apoptosis, and angiogenesis. New findings in muscle from people with PAD included impaired glycolysis and activation of compensatory mechanisms that mitigate and repair damage to DNA, proteins, mitochondria, nerves, and muscle fibers. Unexpectedly, mitochondrial proteins were more abundant in PAD than controls, although in PAD they were disproportionately represented and lacked correlation with mitochondrial respiratory function. These findings suggest that in PAD muscle, impaired mitophagy results in accumulation of fragments of dysfunctional mitochondria. This study identifies new skeletal muscle damage findings in people with PAD suggesting potential targets for to repair tissue and improve mobility in these patients.

Nonstandard Abbreviations and Acronyms

ABI	Ankle-Brachial Index
ETC	electron transport chain
GSEA	Gene Set Enrichment Analysis
ISR	integrated stress response
PAD	peripheral artery disease
TCA	tricarboxylic acid cycle
UPR	unfolded protein response

independent studies, may be targets for interventions to improve skeletal muscle health in PAD.

METHODS

Data Availability

To ensure the transparency and reproducibility of our results, we provide a dedicated GitHub repository (<https://github.com/juliancandia/PAD>) to permanently store the processed datasets and software scripts needed to re-generate the results and figures presented in our study.

Study Population

This study used gastrocnemius muscle biopsy samples from a biobank of muscle biopsies maintained by the Department of Preventive Medicine at Northwestern University's Feinberg School of Medicine (Chicago, IL). The protocol for the collection of such specimens was approved by the Institutional Review Board at Northwestern University. Participants included in the biobank were identified by using postcards mailed to people aged 50 and older living in the Chicago area, advertisements by the Chicago Transit Authority, mailed letters to patients with PAD at Northwestern's Medical Center, a database of participants from prior studies of a coinvestigator (M.M.M.). From the recruitment group, those people with an ankle-brachial index (ABI) indicating a diagnosis of PAD (patients with PAD) and those with normal ABI (non-PAD controls) who donated muscle samples and consented to the use of the collected sample in studies on PAD pathogenesis were eligible for this study. Biopsy specimens from 31 patients with PAD and 29 age- (± 3 years) and sex-matched controls collected at different times between March 25, 2014 and August 20, 2019 were analyzed in this project.

Diagnosis of PAD and Exclusion Criteria

ABI was measured as previously described.⁶ Briefly, a hand-held Doppler probe (Nicolet Vascular Pocket Dop II, Golden, CO) was used to measure systolic pressures at different

locations, twice in the right and left brachial, dorsalis pedis, and posterior tibial arteries using established methods.⁷ The ABI was calculated by dividing the mean systolic blood pressure of the dorsalis pedis and posterior tibial arteries by the mean of the arm systolic pressure. The diagnosis of PAD was established if participants had an ABI<0.90 while controls had an ABI between 0.90 and 1.40. Those with ABIs ranging 0.90 to 0.99 were selected for inclusion if their ABI dropped by at least 20% after 50 consecutive heel rises at a rate of one heel rise per second. Prior work demonstrated that calculating the ABI using the mean of the dorsalis pedis and posterior tibial artery pressures was more closely associated with walking impairment than the alternative method of calculating the ABI using the highest pressure in each extremity.⁷ Patients with PAD with a below-knee or above-knee amputation, as well as participants using a walking aid or wheelchair with severe hearing or visual impairment or with Mini-Mental Status score <23 were excluded. Also excluded were participants with self-reported diabetes because diabetes is well established to have adverse effects on skeletal muscle independently of PAD, a recent cardiovascular event (within the past 3 months), or a rapid poor prognosis because of cancer or other severe medical illness.⁸

Medical History

Medical history, race, education, smoking history, and demographics were obtained by self-report. Coexisting medical conditions were ascertained through medical interview by experienced health practitioners, which included diabetes, cancer, heart failure, angina, myocardial infarction, pulmonary disease, stroke, knee or hip osteoarthritis, and a multimorbidity index calculated as the sum of positive ascertainties. Physical activity level was assessed by a questionnaire derived from the Harvard Alumni Activity Survey.⁹ The physical activity questionnaire asked, "During the last week, how many city-blocks or their equivalent did you walk? Let 12 city blocks equal 1 mile." The 6-minute walking test was performed as detailed elsewhere.¹⁰ Briefly, participants were asked to walk up and down a 30-m hallway for 6 minutes to cover as much distance as possible at their preferred walking speed, and the total distance walked (m) was recorded.

Skeletal Muscle Biopsy Procedures

Skeletal muscle biopsies were taken from the medial head of the gastrocnemius muscle of the leg with lower ABI, as described elsewhere.¹¹ Briefly, an open muscle biopsy was performed under anesthesia with subcutaneous lidocaine. Adipose and subcutaneous tissues were dissected until muscle was identified and biopsied. The muscle specimens collected were cut into portions and immediately frozen in liquid-nitrogen and stored at -80 °C. About 35 to 40 mg were used for this study, half for RNA extraction, and half for protein extraction.

Transcriptomic, Proteomic, Measure of Mitochondrial Respiration, Data Process, and Statistical Analysis

Detailed information is provided in [Supplemental Methods](#).

RESULTS

PAD and Non-PAD Participants and Study Design

Thirty-one patients with PAD, defined by an ABI<0.90 and without chronic limb-threatening ischemia, and 29 people without PAD (ABI>0.90) were included ([Figure S1](#); [Table S1](#)). Participants with diabetes were excluded because diabetes adversely affects skeletal muscle.⁸ PAD participants included a higher proportion of people who were black, had a history of smoking, and were current smokers; major comorbidities were not significantly different between PAD and non-PAD.

Gastrocnemius muscle biopsy specimens dissected from visible fat were used for transcriptomic and proteomic analyses. Differential transcript and protein expression between PAD to non-PAD were performed, followed by Gene Set Enrichment Analysis (GSEA) and Protein Set Enrichment Analyses. Combined transcriptomic and proteomic data were used in coregulation- and network-based analyses on mitochondrial respiratory chain complex proteins and to identify transcript-protein pairs differentially correlated in muscle from PAD versus non-PAD.

Differential Gene Expression and Pathway Enrichment Analysis

Gene expression was quantified for 21 922 annotated genes. An exploratory Principal Component Analysis ([Figure S2A](#)) did not separate PAD and non-PAD, while Partial Least-Squares ([Figure S2B](#)) clearly separated the 2 groups. Of 15 474 transcripts from protein-coding genes, 193 were under-represented and 300 over-represented in PAD versus non-PAD at false discovery rate (FDR)-adjusted P value (P_{adj})<0.05, whereas 29 were under-represented and 45 over-represented at P_{adj} <0.01 ([Figure 1A](#); [Table 1](#) reports names of select significant gene transcripts; [Data S1](#) includes differential gene expression results for all transcripts analyzed). Differentially expressed genes suggest activation of compensatory mechanisms to hypoxia, including lipid degradation, inflammation, mitochondrial metabolism and stress response, angiogenesis, axon protection, fibrosis, response to glucose starvation, and active muscle repair (Notch and WNT).

We did not detect *VEGF165b* mRNA, a splicing variant of *VEGF* mRNA with antiangiogenic properties previously found elevated in muscle from patients with PAD compared to non-PAD.¹² Instead, *VEGF121* mRNA, a splicing variant upregulated by hypoxia with possible pro-angiogenic properties, was over-represented in muscle from PAD participants ([Figure S3C and S3D](#)).¹³

Among 6448 nonprotein-coding transcripts, 125 were differentially expressed between PAD and non-PAD with a P_{adj} <0.05; of these, 6 were under-represented,

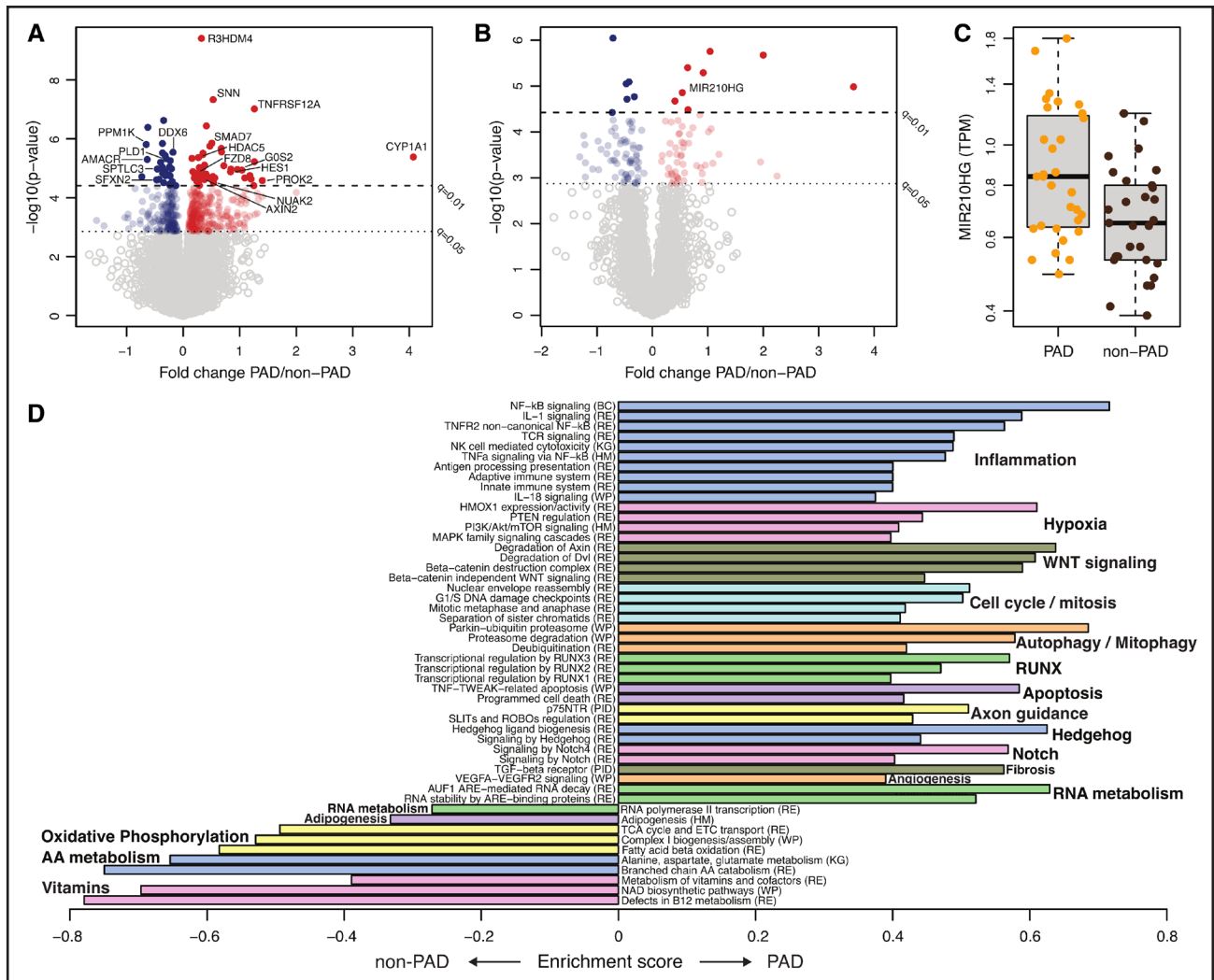


Figure 1. Analysis of transcripts.

A, Volcano plot of differentially expressed protein-coding genes according to effect size and level of significance. **B**, Volcano plot of differentially expressed nonprotein-coding genes according to effect size and level of significance. **C**, Boxplots of *MIR210HG* in peripheral artery disease (PAD) and non-PAD (Wilcoxon test $P < 0.05$). **D**, Select pathways enriched in PAD or non-PAD ($Q < 0.05$; full list in [Data S2](#)). Gene set collections used were Hallmark (HM), Reactome (RE), Kyoto Encyclopedia of Genes and Genomes (KEGG; KG), WikiPathways (WP), Pathway Interaction Database (PID), and BioCarta (BC). ARE indicates adenylate-uridylylate-rich elements; ETC, electron transport chain; IL, interleukin; TCA, tricarboxylic acid cycle; and TNF, tumor necrosis factor.

and 8 over-represented in PAD with a $P_{\text{adj}} < 0.01$ (Figure 1B). This latter group included the long noncoding RNA *miR-210HG* (Figure 1C), which is host to the master microRNA miR-210 that promotes angiogenesis and metabolic adaptation to hypoxia.^{14–17}

Transcript-Level Enrichment Analysis

GSEA revealed 162 biological pathways differentially enriched in PAD relative to non-PAD with a $P_{\text{adj}} < 0.05$ (all 162 pathways with statistics in [Data S2](#)). We focus our description on a parsimonious list of 49 pathways (Figure 1D) selected after excluding duplications; pathways for specific diseases not related to PAD; pathways pointing to general biological mechanisms and pathways with no

known connection with PAD, hypoxia, ischemia-reperfusion or muscle physiology. The selection process that produced the final pathway list is described ([Document S1](#)).

Pathways upregulated in PAD based on GSEA were related to: inflammation, with activation of NF- κ B (nuclear factor κ B), the NLRP3 inflammasome (TLR [Toll-like receptor] activation and upregulation of IL [interleukin]-1 and IL-18), and cell mediated immunity; hypoxia response, including regulation of phosphatase and tensin homolog (PTEN), whose reduction protects cardiac muscle from ischemia¹⁸; WNT, Hedgehog and Notch signaling, which regulate muscle and nerve regeneration, as well as contribute to angiogenesis in response to hypoxia^{19–23}; cell cycle/mitosis, which suggests regenerating muscle tissue; autophagy, with specific mention of the ubiquitin

Table. Differentially Expressed Transcripts and Proteins in PAD vs Non-PAD Selected for Their Role in Hypoxia Response

	Symbol	Role in hypoxia response
mRNA	Under-expressed in PAD ($q < 0.01$)	Biological effect of lower level
	PPM1K	Enhances use of branched-chain α -ketoacids for energy production
	PLD1	Enhances muscle protein degradation through E3-ubiquitin ligases, atrogenes, and the Foxo3 TF
	AMACR	Impairs the β -oxidation of branched-chain lipids in mitochondria
	SPTLC3	Blocks the rate-limiting step in the synthesis of sphingolipids
	SFXN2	Suppresses mitochondrial iron metabolism
	DDX6	Facilitates interferon response and enhances autophagy
	Over-expressed in PAD ($q < 0.01$)	Biological effect of higher level
	SNN	Senses mitochondrial damage and conveys this information to the nucleus
	TNFRSF12A	Promotes the activation of the noncanonical NF- κ B inflammatory pathway
	SMAD7	Inhibits the profibrotic effect of TGF- β
	FZD8	Promotes TGF- β profibrotic signaling
	HDAC5	Upregulates endothelial growth factor (VEGF) gene transcription to enhance angiogenesis and axon regeneration
	G0S2	Protects myocytes from hypoxia, contributes to HIF-1 α stabilization and bioenergetic reprogramming; by inhibiting the activity of ATGL, prevents the accumulation of fatty acids that cannot be processed by β -oxidation
	HES1	Part of Notch signaling that modulates the proliferation-differentiation balance of satellite cells in regenerating muscle
	GDNF	Maintains vitality of motoneurons challenged by hypoxia
	AXIN2	HIF-1 α dependent WNT driven myogenesis during muscle hypoxia
	NUAK2	Enhances survival of myocytes during nutrient deprivation and stress-induced apoptosis
	PROK2	Promotes the proangiogenic effect of VEGF and increases survival of myocytes in models of hindlimb ischemia
Protein	Under-expressed in PAD ($P < 0.05$)	Biological effect of lower level biological effect of lower level
	CD276	Stimulates adaptive immunity by unblocking T-cell activation and proliferation
	PLXC1	Affects stress-induced cytoskeleton rearrangements and secretion of IL-6 and IL-8
	RASA1	Less protection against ischemic stress
	LDHA	Reduces the inter-conversion of pyruvate to L-lactate, impeding the last step of anaerobic glycolysis
Protein	Over-expressed in PAD ($P < 0.05$)	Biological effect of higher level biological effect of lower level
	PTGR2	Modulates inflammation by increasing the conversion of PGE2 to 15-keto-PGE2 (15k-PGE2) in macrophages
	PPM1B	Modulates inflammation by down-regulating NF- κ B and protecting from TNF- α induced necroptosis
	DNAJB6	Participates in the proteostasis response by operating as chaperone
	HspB6	Participates in the proteostasis response by operating as chaperone, exhibits cardioprotective properties during ischemia
	Hsp90A	ATPase-dependent chaperone supports the structural maintenance and function of target proteins during proteostasis stress
	G3BP-2	Triggered by oxidative stress and amino acids starvation, promotes the sequestration of untranslated RNAs into stress granules and stalls translation by promoting the phosphorylation of the subunit α of eIF2 (eIF2 α), thereby protecting cells from apoptosis during stress
	PARP1	Consistent with active DNA repair and NAD $^{+}$ consumption, it is associated with negative effects on mitochondrial biogenesis and mitophagy
	RBBP7	Participates in AMPK-driven mitochondrial biogenesis
	TFAM	Increases mitochondrial transcription and maintenance of mtDNA and biogenesis
	MFR1L	Induces mitochondria fission and mitophagy; protects the vascular system by maintaining a steady-state concentration of peroxynitrite, the product of superoxide radical and nitric oxide
	COX1	A component of cytochrome c oxidase, the last enzyme in the mitochondrial ETC, which catalyzes the reduction of oxygen to water
	RPF2	Involved in mitochondrial biogenesis
	RPL27	Component of the ribosomal 60S subunit

AMACR indicates α -methylacyl-CoA racemase; AMPK, AMP-activated protein kinase; ATGL, adipose triglyceride lipase; AXIN2, axin-related protein; COX1, cytochrome c oxidase subunit 1; DDX6, DEAD-box helicase 6; DNAJB6, DnaJ homolog subfamily B member 6; ETC, electron transport chain; Foxo3 TF, Forkhead Box O3 Transcription Factor; FZD8, Frizzled receptor 8; G0S2, G0/G1 switch gene 2; G3BP-2, Ras GTPase-activating protein-binding protein 2; GDNF, glial cell-derived neurotrophic factor; HDAC5, histone deacetylase 5; HES1, Hes family BHLH transcription factor 1; HIF1 α , hypoxia-inducible factor 1 α ; Hsp90A, heat shock protein HSP 90- α ; HspB6, heat shock protein β -6; IL, interleukin; LDHA, lactate dehydrogenase A; MFR1L, mitochondrial fission regulator 1-like; mtDNA, mitochondrial DNA; NF- κ B, nuclear factor κ B; NUAK2, NUAK family kinase 2; PAD, peripheral artery disease; PARP1, poly (ADP-ribose) polymerase 1; PGE2, prostaglandin E2; PLD1, phospholipase D1; PLXC1, virus-encoded semaphorin protein receptor; PPM1B, protein phosphatase 1 B; PPM1K, phosphatase 1K; PROK2, prokineticin 2; PTGR2, prostaglandin reductase 2; RASA1, RAS P21 protein activator 1; RBBP7, histone-binding protein RBBP7; RPF2, ribosome production factor 2; RPL27, ribosomal protein L27; SFXN2, sideroflexin 2; SMAD7, SMAD family member 7; SNN, stannin; SPTLC3, serine palmitoyltransferase; TFAM, transcription factor A, mitochondrial; TGF, transforming growth factor; and TNFRSF12A, tumor necrosis factor receptor superfamily member 12A.

proteasome system that regulates mitochondria quality control; RUNX, a family of DNA-binding proteins with pleiotropic activities, including DNA repair and the inhibition of the atrophic program induced by muscle denervation²⁴; apoptosis, which has been demonstrated in PAD muscle²⁵; axon guidance, with an emphasis on Slit/Robo signaling, which play a role in axon pathfinding in regeneration of injured peripheral nerves²⁶; fibrosis induced through TGF (transforming growth factor)- β receptors²⁷; angiogenesis (VEGFA signaling); and RNA metabolism, with an emphasis on ARE (adenylate-uridylylate-rich elements)-mediated RNA decay, a quality control system that fine-tunes gene expression during cellular stresses such as amino acid deprivation and hypoxia.²⁸

Pathways downregulated in PAD compared to non-PAD were related to oxidative phosphorylation, specifically fatty acid β -oxidation, tricarboxylic acid cycle (TCA), ETC (electron transport chain), and complex I biogenesis; adipogenesis, which is typically downregulated during hypoxia; amino acid metabolism, including the catabolism of branch chain amino acids that contribute to mitochondrial biogenesis by enhancing SIRT1 activity²⁹; and vitamin metabolism, including deficiency in the NAD⁺ biosynthetic pathway that may negatively affect mitochondrial function.³⁰

Differential Protein Expression and Pathway Enrichment Analysis

An exploratory Principal Component Analysis (Figure 2C) did not separate PAD and non-PAD, whereas Partial Least-Squares (Figure 2D) clearly separated the 2 groups. Comparing PAD to non-PAD, 16 proteins (out of 6324) were under-represented and 27 over-represented with a $P < 0.01$, whereas 94 were under-represented and 173 over-represented with a $P < 0.05$ (Figure 2A; Table 1 reports gene names of selected significant proteins; Data S3 includes Differential Protein Expression results for all proteins). Only CD276 (which regulates adipogenesis and controls aerobic glycolysis through reactive oxygen species-mediated stabilization of HIF1 α [hypoxia-inducible factor 1 α]³¹) was over-represented with $P_{\text{adj}} < 0.05$.

Under-represented proteins in PAD compared to non-PAD suggested increased inflammation and downregulation of glycolysis. Over-represented proteins in PAD are consistent with activated inflammation, proteostasis (heat shock response), integrated stress response (ISR) to amino acid starvation,³² DNA repair (which may reduce NAD⁺ availability), translation, and mitochondrial metabolism and biogenesis. Of note, 28 mitochondrial proteins were differentially expressed between participants with and without PAD: 14 of these were from the inner membrane (HEMH, NDUB9, COX5B, NDUA8, RM50, COX1, RM54, RM27, SDHA, TIMB8, RT11, ATP6, CHCH1, SCO2), 4 from the outer membrane (MRF1L, GLPK3, GAL3A, TOM40), and 10 from the mitochondrial matrix

(BCDO2, P5CR2, HMCS2, SCOT1, PYR2, MMSA, TFAM, CASQ1, SYKM, PYRD; Data S3).

A heatmap (Z values) for the 96 mitochondrial ETC-related is shown in (Figure 2B). On average, ETC-related proteins were over-represented in PAD relative to non-PAD. There was substantial heterogeneity, which is emphasized by ordering participants from low to high overall respiratory protein content in non-PAD and from high to low overall respiratory protein content in PAD (Figure 2B). A heatmap representation of the mRNAs encoding the same mitochondrial respiratory proteins similarly ordered as Figure 2B also showed similar heterogeneity among the study participants (Figure S4A). However, ordering participants exactly as in Figure 2B revealed that the heterogeneity in transcripts and proteins is not due to the same individuals, indicating a dissociation between mRNA and protein levels (Figure S4B), possibly due to mRNA turnover, preferential translation, or regulated protein stability. Eighteen proteins related to ribosome structure or function were differentially expressed between PAD and non-PAD: 7 of these were cytoplasmic ribosome proteins (RPL8, RPL17, RPL18A, RPL27, RPL31, RPL32, RPS20), 5 were mitochondrial ribosome proteins (MRPL27, MRPL50, MRPL54, MRPS11, CHCHD1), and 6 were proteins involved in ribosomal biogenesis (URB2, RIOK2, NPM1, NCL, WDR12, RPF2; Figure 2C; Data S3).

Protein Set Enrichment Analyses support results of the Differential Protein Expression analysis, with 69 significantly enriched pathways between PAD and non-PAD with a $P_{\text{adj}} < 0.05$ (all 69 pathways with statistics shown in Data S4). Thirty pathways were selected for presentation in Figure 2D based on the same criteria described for the GSEA (Document S1).

Significantly enriched pathways upregulated in PAD pertain to WNT signaling (already highlighted by the GSEA); mitochondria TCA, ETC, complex I biogenesis and complex IV assemblages; translation (rRNA processing as well as transfer RNA [tRNA] aminoacylation, translation initiation, and elongation); RNA processing and splicing (including nonsense-mediated decay that together with ARE-mediated RNA decay was enriched by GSEA); response of EIF2AK4 to amino acid deficiency and cellular response to starvation are part of the ISR; axon guidance with emphasis on Slit/Robo signaling already enriched in GSEA. Inflammation was also upregulated in PAD. The pathways, Reactome cell surface interaction at the vascular wall related to platelet and leukocyte interactions through the endothelium and Reactome Fc Epsilon receptor FCER1 signaling, related to mast cells' role in hypersensitivity and allergic reactions were downregulated in PAD. Vascular endothelium may be so damaged in PAD that surface interactions are deranged. Allergic reactions are likely not relevant to PAD.

The pathway hypoxia was enriched in participants without PAD compared to PAD but of the 20 leading

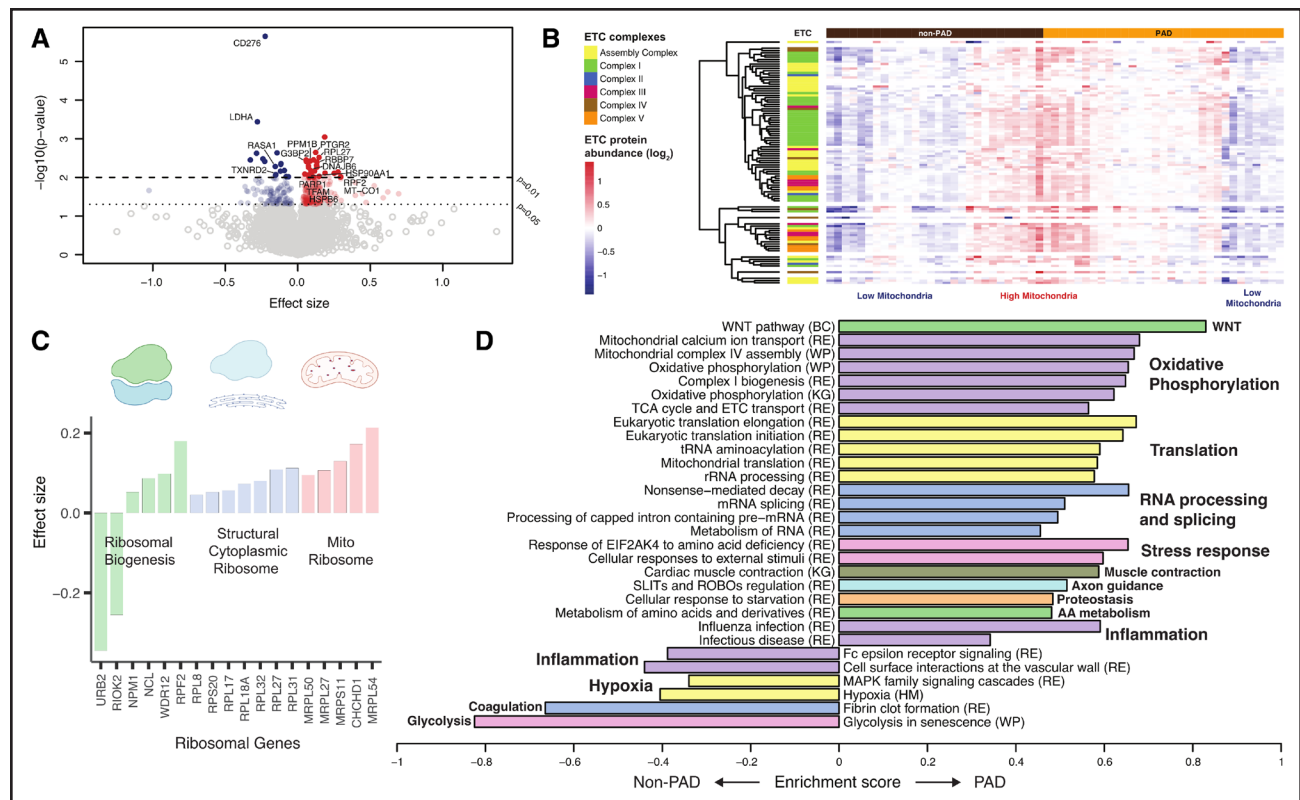


Figure 2. Analysis of proteins.

A, Volcano plot showing proteins over- and under-represented in peripheral artery disease (PAD) compared to non-PAD according to effect size and level of significance. **B**, Heatmap of 96 mitochondrial respiratory proteins in PAD and non-PAD. Participants are ordered according to average protein level in ascending (non-PAD) and descending (PAD) order. **C**, Bar chart showing ribosomal-related proteins over- and under-represented in PAD compared to non-PAD. **D**, Selected pathways enriched in PAD or non-PAD ($Q < 0.05$; full list in [Data S4](#)). Gene set collections used were: Hallmark (HM), Reactome (RE), Kyoto Encyclopedia of Genes and Genomes (KEGG; KG), WikiPathways (WP), Pathway Interaction Database (PID), and BioCarta (BC). ETC indicates electron transport chain.

proteins, 14 were related to glycolysis or glycogenolysis ([Data S4](#), [Figure S5](#)). These findings are consistent with impaired skeletal muscle glucose uptake measured by positron emission tomography (PET) in patients with PAD.³³

Consistent with previous evidence of increased abundance of mitochondrial DNA with greater lower extremity ischemia in PAD,¹¹ and in contrast to the transcriptomic analysis, Protein Set Enrichment Analyses suggest that mitochondrial proteins were over-represented in PAD relative to non-PAD muscle, including TCA cycle and the ETC proteins, as well as proteins involved in mitochondrial translation, biogenesis (complexes I and IV), and calcium channels.

Mismatch Between Transcriptomic and Proteomic Results of Mitochondrial Proteins in PAD

Some pathways were enriched in the same direction in the transcriptomic and proteomic analyses while others displayed inverse relationships, particularly for mitochondrial proteins. Of the significant transcripts-proteins

differentially expressed both in differential gene expression and Differential Protein Expression, 17 mRNAs were less abundant while the encoded proteins were more abundant in PAD compared to non-PAD; this group included mitochondria TFAM (transcription factor A), a key mitochondrial transcription factor that contributes to mtDNA replication, transcription and repair ([Table S3](#)).³⁴

The imbalance between omics in our analysis and past reports of impaired mitochondrial activity in PAD suggests that in PAD muscle part of mitochondrial proteins may not be part of complete functional units but rather represent fragments of respiratory complexes not removed because of defective mitophagy.³ Excess mitochondrial proteins may inhibit transcription of their related genes explaining the discrepancy between transcripts and proteins detected in this study. To test this hypothesis, a coregulation network analysis of the 96 mitochondrial ETC proteins was performed in PAD ([Figure 3A](#)) and non-PAD ([Figure 3B](#)). Respiratory proteins were more loosely connected in PAD than non-PAD subjects. The PAD network showed a greater range of protein abundance for the same complexes, with the average closeness centrality significantly lower compared to non-PAD

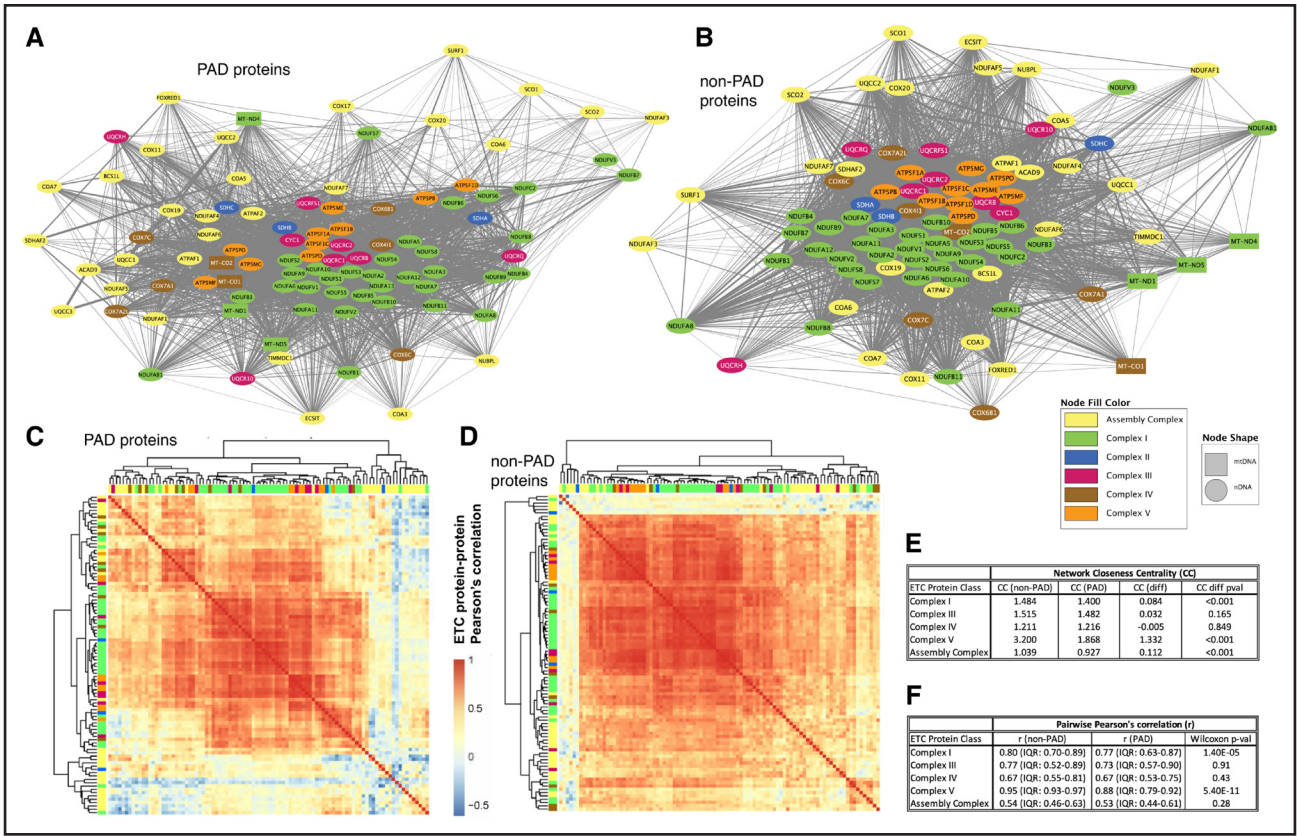


Figure 3. Coexpression analysis of 96 mitochondrial respiratory proteins in peripheral artery disease (PAD) and non-PAD subjects. **A** and **B**, Undirected, weighted networks for each group, where nodes represent proteins (colored by protein complex) and links represent protein-protein correlations (whose widths increase monotonically with correlation). Different node shapes indicate whether proteins are encoded in nuclear or mitochondrial DNA. **C** and **D**, Hierarchically clustered heatmap of Pearson correlation matrix for ETC (electron transport chain) proteins in each cohort. **E**, Network closeness centrality measurements comparing PAD and non-PAD for each ETC complex. **F**, Pearson correlation measurements comparing PAD and non-PAD for each ETC complex.

for complexes I and V, as well as for the assembly complex (Figure 3E). Consistent with these findings, the PAD cocorrelation matrix showed weaker associations between respiratory proteins (Figure 3C) than in non-PAD (Figure 3D), where most ETC proteins appeared highly correlated, suggesting that in non-PAD participants, but not in PAD participants, respiratory proteins were in the optimal stoichiometric proportions to be parts of complete respiratory chains. These observations were further confirmed by statistical comparisons of pairwise correlation for each protein complex (Figure 3F). In contrast, an analogous network analysis using the mRNAs instead of the proteins for the same ETC system (Figure S6A and S6B) showed highly coexpressed mRNAs in both PAD and non-PAD (Figure S6C).

To verify the above findings, we used an Agilent Seahorse XFe Flux Analyzer to measure the activity of mitochondrial complexes in the 45 participants from whom we had remaining frozen muscle specimens.³⁵ We obtained reliable signals for complex II (non-PAD, n=17; PAD, n=21) and complex IV (non-PAD n=14 and PAD n=15) activities (Figure S8A). Characteristics of the 39 subjects with data on complex II or complex IV activity

are shown in Table S2. Similar to the whole study sample, PAD participants included in this subsample were more frequently black, had a history of smoking, and were current smokers. For both complex ii and complex IV, the differences in mean activity between PAD and non-PAD were not statistically significant (Figure S8B). However, complex II and complex IV activities were significantly correlated in non-PAD participants (Spearman $\rho=0.66$, $P=0.001$) but not in PAD participants ($\rho=0.52$, $P=0.517$; Figure 4A). We next constructed mitochondria respiratory proteomic scores as the mean expression of all mitochondrial respiratory proteins (mitochondria protein score), complex II proteins (complex II protein score), or complex IV proteins (complex IV protein score) and then computed Spearman rank order correlations among these scores and the activities of complex II and complex IV (Figure 4B). In non-PAD participants, the mitochondria protein score was positively correlated with complex II activity ($\rho=0.46$; $P=0.003$) and was close to statistical significance with respect to complex IV activity ($\rho=0.25$; $P=0.087$); in PAD participants, the score was not correlated with the activity of either complex II ($\rho=0.26$; $P=0.173$) or complex IV ($\rho=0.22$;

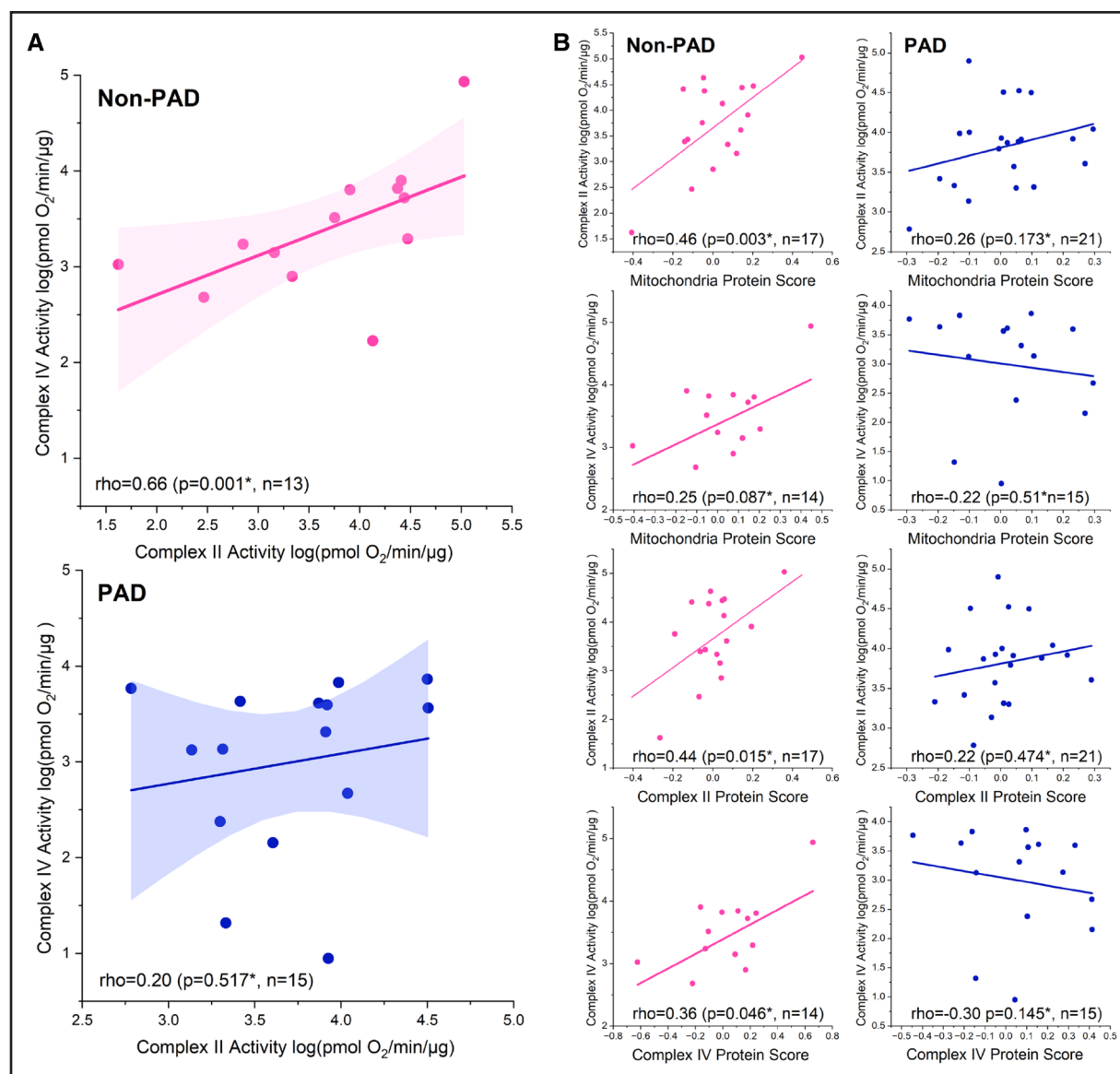


Figure 4. Complex II and complex IV activity in peripheral artery disease (PAD) and non-PAD subjects.

A, Scatterplots of correlations between the activity of complex II and complex IV in non-PAD and PAD participants. Shaded areas around the regression line indicate 95% CIs. B, Scatterplot of the mitochondria protein score, complex II protein score and complex IV protein score with complex II and complex IV activity assessed by respirometry. ρ values are Spearman correlation coefficients and P values are from robust linear regression.

$P=0.510$). Similarly, the complex II protein score was significantly correlated with complex II activity in non-PAD participants ($\rho=0.44$; $P=0.015$) but not in PAD participants ($\rho=0.18$; $P=0.43$); and the complex IV protein score was significantly correlated with complex IV activity in non-PAD participants ($\rho=0.36$; $P=0.046$) but not in PAD participants ($\rho=-0.30$; $P=0.145$; Figure 4B). These findings support the hypothesis that mitochondrial proteins accumulate in PAD muscle, perhaps due to impaired mitophagy.³ We found no significant correlation between complex II activity, complex IV activity, and mitochondria protein score with ABI or 6-minute walk performance in either non-PAD or PAD participants (Figures S7 and S9). In contrast, the relationship

between the ETC proteomic score and the self-reported city blocks walked was statistically significant in the non-PAD participants ($P=0.008$) but not in the PAD participants ($P=0.17$), supporting the hypothesis that a portion of the mitochondrial ETC proteins in PAD muscle are not functional.

Disconnect Between Levels of mRNA and Proteins in PAD Muscle

The lack of correlation between the abundance of mRNAs and encoded proteins that may arise from differences in the production or processing of mRNA (eg, transcription, export to cytoplasm, and stability) or

proteins (eg, translation and degradation), a feature consistent with activation of the ISR suggested by both GSEA and Protein Set Enrichment Analyses. To further uncover mechanisms at the interface between transcription and translation in PAD, we analyzed differences in pairwise transcript-protein correlations between PAD and non-PAD for the 5851 genes with expression and protein data available (Data S5). Figure 5A shows a scatterplot for each mRNA-protein pair correlations in PAD (y axis) plotted against non-PAD (x axis). Colored symbols identify transcript-protein pairs for which: (1) the correlation between transcripts and proteins is significant ($P < 0.05$) either in PAD or non-PAD participants, and (2) the correlation in PAD lies outside the 95% CI range defined by the correlation in non-PAD samples. Colored circles (128) in the upper left quadrant indicate higher correlation of mRNA and corresponding proteins in PAD relative to non-PAD (Figure 5A). In this group, RBX1 (ring-box 1) is part of the complex that regulates the stability of HIF-1 α ³⁶; UQCC3 (ubiquinol-cytochrome C reductase complex assembly factor 3) is a mitochondrial protein that drives cellular adaptation to hypoxia³⁷; and the hypoxia-inducible protein G0/G1 switch gene 2 (G0S2, also significantly upregulated in PAD, see

Figure 1A) enhances oxidative phosphorylation, protects cardiomyocytes from hypoxia and promotes HIF-1 α stabilization.^{38,39} In an enrichment analysis of these 128 selected pairs, the top enriched gene set was microtubulins, with gamma-tubulin complex components 2 and 3 (TUBGCP2, TUBGCP3) as the most contributing genes. Microtubule remodeling is integral to the hypoxia response, as it facilitates the translocation of HIF-1 α to the nucleus and directs protein localization to other organelles.^{40,41} Remodeling of the cytoskeleton was suggested by downregulation of PLXC1 (virus-encoded semaphorin protein receptor) that commonly occurs during hypoxia-driven stress.⁴² Recent evidence suggests that in the ISR, microtubule organization is remodeled to contribute to the unfolded protein response (UPR) following proteotoxic stress.⁴³ The second most enriched gene set was extracellular organelle ($P = 0.0051$), which encompasses extracellular vesicles (EVs; blue circles, significant genes shown in Figure 5B).

The 195 colored circles in the lower right quadrant represent mRNA-protein pairs that had higher correlation in non-PAD relative to PAD samples (Figure 5A). This group was enriched for citric acid cycle and respiratory electron transport ($P = 0.0002$), processing of SMDT1

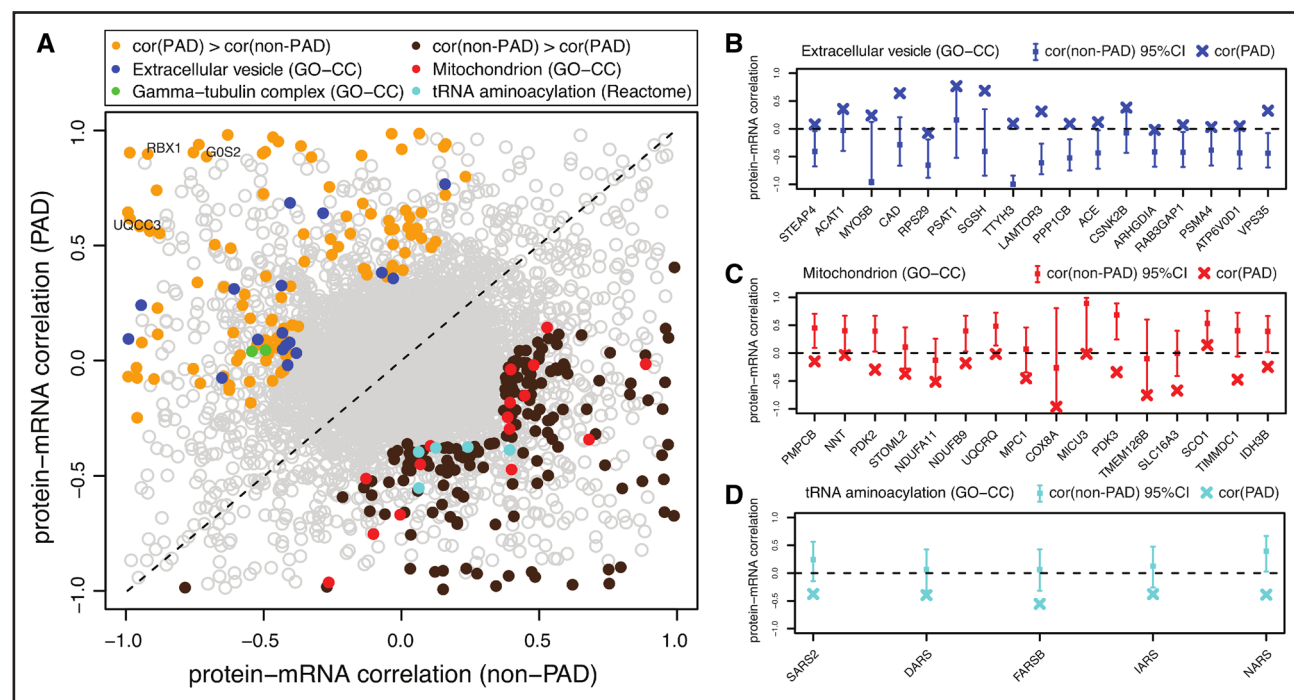


Figure 5. mRNA-protein correlations in peripheral artery disease (PAD) and non-PAD.

A, Scatterplot of mRNA-protein correlations in PAD and non-PAD limited to the 5851 genes for which both mRNA and protein expression was available. Colored symbols indicate (1) significant correlations ($P < 0.05$) between mRNAs and proteins either in patients with PAD or non-PAD controls, and (2) correlation coefficients in PAD outside the 95% CI defined for correlations in non-PAD samples. Other mRNA-protein correlation pairs are indicated by gray empty circles. Because of missing data, some pairs have high but still nonsignificant correlation values in both patients with PAD and non-PAD controls. The filled circles on the **upper-left** quadrant represent transcript-protein pairs that have higher correlation in PAD than in controls. Enriched pathways by Gene Set Enrichment Analysis for off-diagonal genes are shown on the **right**, namely gamma-tubulin complex (green circles), extracellular vesicles (blue circles and **B** for individual genes in the pathway), mitochondrion (red circles and **C**), and tRNA aminoacylation (cyan symbols and **D**). CI indicates confidence interval; cor, Pearson's correlation; GO-CC, gene ontology-cellular component; and tRNA, transfer RNA.

($P=0.0012$), and mitochondrial calcium ion transport ($P=0.00298$), which collectively encompass primarily mitochondrial proteins (red circles, significant genes in Figure 5C). There was also enrichment for the cytosolic tRNA aminoacylation pathway ($P=0.00089$), an early step in translation initiation⁴⁴ (cyan circles, Figure 5D). These findings suggest that PAD causes rearrangement of the cytoskeleton and modulates EV production.

DISCUSSION

PAD is a debilitating disease for which no highly effective drug treatment exists.⁵ Supervised and home-based walking exercises improve mobility in patients with PAD, but progress is hampered by poor adherence, lack of durability, and heterogeneity of effectiveness.^{45,46} A better understanding of mechanisms of skeletal muscle damage in PAD may help identify new therapeutic opportunities. This study summarizes biological pathways activated in PAD muscle, confirms some expected hypoxia responses, but identifies novel mechanisms that are putative new intervention targets for PAD.⁴⁷

Our findings are consistent with a coordinated network of hypoxia responses in PAD muscle.⁴⁸ The modulation of the PTEN-PI3K pathway⁴⁹ and MAPK (mitogen-activated protein kinase) signaling have been shown to protect cardiomyocytes and promote repair after myocardial infarction.⁵⁰ Increased TGF- β signaling was previously observed in hypoxic skeletal muscle of patients with PAD.^{4,51} Continuous repair of muscle and nerve tissues damaged by chronic hypoxia or ischemia-reperfusion requires the coordinated action of the Notch, WNT, and Hedgehog pathways by coregulation of shared target genes, such as HES1 (Hes family BHLH transcription factor 1).^{21,52} Interestingly, both the transcriptomic and proteomic analysis showed activation of WNT that may also contribute to fibrosis.^{53,54} Angiogenesis is upregulation of VEGF1, the Hedgehog pathway, interactions with the RUNX (RUNX-related transcription factor) 1 to 3, and miR-210.^{55–58} We found no evidence of differentially expressed noncoding RNAs previously described in the literature besides miR-210, possibly because our RNA library preparation did not specifically enrich for microRNAs and long noncoding RNAs (lncRNAs).

Results are consistent with an inflammatory environment in PAD muscle, although specific cytokines were not detected in the proteomic analysis, possibly because liquid chromatography mass spectrometry (LC-MS) cannot detect low-abundance proteins such as cytokines.⁵⁹ The transcriptomic analysis also highlights damage to mitochondria, resulting, which is likely responsible for NF- κ B activation and increased expression of proinflammatory genes, such as TNF (tumor necrosis factor)- α and IL-6, which in turn activate immune cells, such as lymphocytes and natural killer cells.⁶⁰ Damaged mitochondria release oxidized mitochondrial DNA, free radicals, and cardiolipin

into the cytosol,⁶¹ activating the NLRP3 which was witnessed by upregulation of IL-1 β and IL-18.⁶² Enhanced antigen presentation in PAD is suggested by GSEA and also by higher levels of S1008A and S1009A mRNAs that are biomarkers of macrophage activation.¹²

There was discrepancy between transcriptomic and proteomic analyses concerning mitochondrial genes. The transcriptome showed a negative enrichment of multiple mitochondrial pathways, including those involved in respiration (ETC), energetic metabolism (TCA), and respiratory complex biogenesis, whereas the proteomic analysis shows positive enrichment of the same pathways, with some overlap at the gene level (Figures 1D and 2D). A possible interpretation of these findings is that low ATP availability in lower extremity muscle of people with PAD stimulates mitochondrial biogenesis and enhances mitochondria protein translation. However, damaged and fragmented mitochondria are not removed because of defective mitophagy, leading to the accumulation of mitochondrial proteins that inhibit further transcription of the genes encoding these proteins. Such findings are consistent with literature indicating that mitochondrial mass is higher in the skeletal muscle of PAD-affected individuals, despite evidence of either lower or comparable mitochondrial function.^{11,63,64} The poorer correlation between mitochondria respiratory chain proteins in PAD than in controls (Figure 3) indicates these proteins in PAD lack the stoichiometric proportions required to be part of complete functional units, possibly because they are not removed by defective mitophagy.³ This interpretation is also consistent with previous findings that the correlations between muscle mitochondria DNA copy number and 6-minute walk performance is positive in PAD but negative in non-PAD.³¹ Finally, supporting this hypothesis, the abundance of mitochondrial respiratory proteins correlated significantly with complex II and complex IV activity in non-PAD participants but not in PAD participants. Also, the average abundance of mitochondrial respiratory proteins significantly correlated with self-reported mobility in non-PAD but not in PAD (Figure S7). Testing whether directly enhancing autophagy, via pharmacological mechanisms or by increased perfusion or exercise, restores ATP production and improves performance in patients with PAD deserves further investigation.

The proteomic analysis found that most glycolytic and glycogenolytic enzymes, some of which catalyze rate-limiting pathway steps, such as hexokinase and pyruvate kinase, were under-represented in PAD muscle (Figure 5D). This finding is puzzling because anaerobic glycolysis is an essential alternative source of ATP during hypoxia when oxidative phosphorylation is impaired.⁶⁵ Also, down-regulated glycolytic enzymes contrast with the upregulation of miR-210 and *GOS2* mRNA that are thought to enhance glycolysis. Yet, these findings are consistent with previous reports of impaired skeletal muscle glucose

uptake in patients with PAD.³³ Our results should be confirmed by performing the Seahorse extracellular acidification rate assay on freshly collected biopsy samples but suggest that interventions that enhance glycolysis may be beneficial in patients with PAD. The metabolism of branched-chain amino acids that takes place in mitochondria was also impaired in PAD.^{29,66} Thus, branched-chain amino acids cannot be catabolized to acetyl-CoA or succinyl-CoA and fuel the TCA cycle, suggesting additional metabolic features that can be targeted therapeutically.

Both the proteomic and transcriptomic analyses point to active regeneration (cell cycle/mitosis). Nerve damage in PAD has been observed in previous studies, and our findings point to a role of the secreted glycoprotein SLIT and the receptors ROBOS in axon pathfinding and neuronal migration during peripheral nerve repair.⁶⁷

Enriched pathways in PAD suggest the activation of regulatory mechanisms at the interface of mRNA and protein abundance. There was enrichment of transcripts and proteins implicated in the rapid degradation of mRNAs due to the presence of AREs in their 3' untranslated regions, a response mechanism to different stresses, including hypoxia,⁶⁸ along with an upregulation of the minor and major splicing pathways and higher nonsense-mediated RNA decay, a quality control mechanism that eliminates abnormal mRNAs that arise during inaccurate splicing.^{69,70} Splicing and RNA processing are considered resilience strategies in response to ATP scarcity due to mitochondrial dysfunction.⁷¹ Additionally, there was evidence of activation of the UPR, a proteostatic response to accumulated unfolded proteins in the endoplasmic reticulum that aims to restore normal folding and trafficking and to suppress translation to avoid overloading the folding system.⁷² Nonsense-mediated RNA decay also contributes to the UPR by promoting the degradation of aberrant transcripts that carry premature STOP codons and by increasing the decay of critical UPR transcripts.⁷³ The ISR inhibits general protein translation while enhancing the translation of proteins that promote cell survival and reprogram gene expression and signaling pathways to restore protein homeostasis (such as HSF1, HSPB6, and HS90A found upregulated in PAD) and enhance autophagy flux.^{74,75} Overall, these findings suggest that the ISR and UPR deserve further investigation in PAD and may be promising therapeutic targets, as proposed for central nervous system (CNS) injuries.⁷⁶

Genes with the large mRNA-protein expression differences were the gamma-tubulin system and EVs. It has been previously shown that desmin, a major subunit of the cytoskeleton, is aberrant in PAD gastrocnemius and correlates with the degree of pathology and functional impairment.⁷⁷ However, the role of the microtubules, another major component of the myofiber cytoskeleton, a likely factor in the ISR, has not been investigated in PAD. Enhanced production of EVs triggered by hypoxia has been extensively studied in cancer and coronary artery disease, but very little in

PAD.^{78,79} EVs produced by hypoxic mesenchymal cells contain miR-210 (which was found elevated in this study) and protect myocardial cells during overt ischemia.⁸⁰ The study of peripheral EVs may therefore open a new window into understanding PAD myopathy.

The only work that is partially comparable to our study is the RNA sequencing (RNA-seq) analysis of gastrocnemius biopsies, where Ryan et al reported 397 differentially expressed genes between 20 patients with PAD (with no critical limb ischemia) and 15 controls.⁸¹ In this study, information was available for 384 of the aforementioned 397 genes, with 176 over-represented and 208 under-represented in PAD. Here, 113 (~30%) of the 384 were differentially expressed ($P < 0.05$) in the same direction (up or down) as reported by Ryan (Figure S3E, Data S6). The overlap of results is remarkable, and the partial discrepancy can be explained by differences in study population characteristics and the sequencing method. Ryan's data were reanalyzed by Guangzhi Cong et al,⁸² who described an increased abundance of fibrosis in PAD muscle compared to controls only for the subgroup of participants with critical limb ischemia. These findings contrast somewhat with the enrichment of the TGF- β pathway (and upregulation of SMAD mediators) in our study, suggesting a fibrotic process already active in PAD without critical limb ischemia, in line with previous literature.²⁷ Newman et al⁸³ compared whole blood gene expression in 42 patients with PAD and 29 sex-matched controls and reported 127 differentially expressed genes, 94 of which were quantified in our study. Of these, only 7 were significant and consistent between the two datasets, perhaps reflecting the different nature of the biospecimens (Figure S3F, Data S7). Saini et al⁸⁴ performed RNA-seq analysis in gastrocnemius muscle in 6 individuals free of PAD compared to 7 patients with PAD who later underwent mobility decline and 7 patients with PAD who remained functionally stable. They found few differences between the non-PAD participants and the participants with PAD who did not experience progression of PAD, defined by either ABI decline or 6-minute walk decline. Some of the over-represented pathways in the decliners compared to controls, that is, inflammation (S100A8, S100A9), TGF- β , and NOTCH signaling (HES1), were also found significantly over-represented in our study. Conversely, whereas Saini and colleagues detected enrichment of growth/differentiation factors, activin, and cellular senescence, we did not. Differences may be explained by the small sample size or the substantial differences in the RNA-seq depth between the current work and Saini et al's.⁸⁴

The study has limitations. Because of the cross-sectional nature of the analysis, our findings should be considered as hypotheses generating and do not allow causal inferences. We used a GSEA method to help interpreting the differentially expressed genes and proteins between PAD and non-PAD,⁸⁵ but enrichment methods

have limitations, such as enrichment bias dependent on pathway size or effects of overlaps between pathways.⁸⁶ We did not perform direct measures of major mediators of the hypoxia response, such as HIF-1 α , AMPK (AMP-activated protein kinase), and PGC-1 α . The sample size was relatively small and did not allow subgroup analysis. For example, we cannot exclude that smoking or unmeasured comorbidities may have contributed to differences seen between groups.⁸⁷ The findings may not be generalizable to people with PAD who have diabetes or critical limb ischemia, as these subjects were excluded from our analyses. Despite these limitations, we provided a comprehensive picture of the myopathy associated with PAD, including the many biologic pathways that are activated in hypoxic muscle, some of which have not been previously described in PAD and may indicate new intervention targets to limit lower extremity muscle damage in PAD and their functional consequences on mobility.

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Disclosures

None.

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